

Animesh Biswas

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EDUCATION

Kalinga Institute of Industrial Technology (KIIT), Bhubaneswar, Odisha

B. Tech, Electronics and Telecommunication Engineering

CGPA: 8.23/10.0 | Expected Graduation: May 2027

Jamshedpur Public School, Jamshedpur, Jharkhand

Higher Secondary Certificate (CBSE Class XII) | Percentage: 66% | Completed: March 2023

Secondary School Certificate (CBSE Class X) | Percentage: 85% | Completed: March 2021

RESEARCH PUBLICATIONS

Design and Implementation of a High-Performance 16-Bit DSP ALU Using Six-Stage Pipelining | *IEEE IC2PCT 2026 (Published)*

Tools: Xilinx Vivado Design Suite, Verilog HDL, Digilent Nexys 4 DDR (Xilinx Artix-7)

- Developed a 6-stage pipelined architecture on Nexys 4 DDR, optimizing throughput by explicitly inferring DSP48E1 hard macros for multiplication.
- Achieved 100 MHz operational frequency and low dynamic power (25mW) by implementing latency balancing and register retiming strategies.
- Code Repository: github.com/animesh9364/DSP-ALU-6-Stage-Pipelined-VLSI

INDUSTRIAL EXPERIENCE

Industrial Trainee - EV Powertrain Architecture | *Tata Motors Limited (June 2026 – July 2026)*

- Analyzed high-voltage network architecture (300V–775VDC), MCU inverter design, and 453 kWh LFP battery liquid thermal management systems for heavy commercial electric vehicles.
- Observed and documented the physical component fittings, chassis assembly, and high-voltage component packaging of the Tata Prima E.28K EV platform, including e-axle layout, traction motor mounting, and battery liquid cooling loops.
- Evaluated High-Voltage Interlock Loop (HVIL) safety diagnostics, galvanic isolation barriers, and dual-channel SAE J1939 CAN communication networks.

PROJECTS

Power System Fault Classification Using Intelligent Machine Learning with Hardware Implementation | *KIIT (Ongoing - 7th to 8th Sem)*

Tools: Python, PyTorch, Raspberry Pi 5, ADS1256 ADC, Scikit-learn, MATLAB

- Designed a two-stage ML architecture using Decision Trees and a late-fusion ensemble (KAN, SNN, LNN) for classifying multi-phase grid faults from 6-channel input datasets (V_a , V_b , V_c , I_a , I_b , I_c), evaluating software performance across accuracy, latency, confusion matrices, and loss curves.
- Developing the ongoing hardware integration framework with Raspberry Pi 5, an 8-channel ADS1256 ADC, and isolation transformers to enable real-time hardware inference upon completion of the lab generator testbench setup.

Live AI Quantitative Trading System | *Personal Project (Ongoing - 2026 to 2027)*

Tools: Python, Pine Script v6, Scikit-learn, Alpaca API, Pandas

- Developed an automated algorithmic trading pipeline integrating custom Pine Script v6 indicators with machine learning models for real-time market trend forecasting.
- Implemented automated order execution and risk controls via REST APIs, evaluating strategy performance through historical backtesting and live latency monitoring.

TECHNICAL SKILLS

Languages: C++, C, Python, Verilog

Software: KiCad, Xilinx Vivado, VS Code, Arduino IDE, MATLAB

Hardware: Raspberry Pi 5, ESP32, FPGA, PCB Design & Prototyping

Domains: EV Power Electronics, MCU Architecture, Embedded Systems, Machine Learning, Signal Conditioning